

**EGERTON**



**UNIVERSITY**

**UNIVERSITY EXAMINATIONS**

**REGULAR/SSP -NJORO CAMPUS**

**SECOND SEMESTER, 2024/2025. ACADEMIC YEAR**

**FIRST YEAR EXAMINATION FOR THE DEGREE OF BACHELOR OF COMMERCE,**

**BACHELOR OF PROCUREMENT AND SUPPLY CHAIN MANAGEMENT AND**

**BACHELOR OF ENTREPRENEURSHIP AND SMALL BUSINESS MANAGEMENT**

**CBIS 151: FUNDAMENTALS OF INFORMATION SYSTEMS**

**STREAM: BCOM, BPSM & ENTRE**

**TIME: 2 HRS**

**EXAMINATION SESSION: APRIL**

**YEAR: 2025**

**INSTRUCTIONS**

- (i) Answer question ONE and any other TWO questions
- (ii) Each question should start on a new page
- (iii) Do not write on the question paper

**QUESTION ONE**

**(30 Marks)**

a) Explain the following terms as used in Business Information Systems.

- i) Entity-relationship (E-R) diagrams **(2 Marks)**
- ii) Business intelligence (BI) **(2 Marks)**
- iii) Database management system (DBMS) **(2 Marks)**
- iv) Global positioning systems (GPS) **(2 Marks)**

b) Waterfall model is a linear and sequential framework for system development and follows the basic phases of SDLC model. Highlight three benefits of this model. **(3 Marks)**

c) Explain how information systems apply competitive strategies in business. **(3 Marks)**

d) Explain how organizations are using Voice over Internet Protocol (VOIP) in their business operations today. **(3 Marks)**

- c) Data security is where protective measures are put in place within an organization to keep unauthorized access out of computers, websites and databases. Identify five steps Egerton University can pursue to achieve data security. **(5 Marks)**
- f) Managers in organization benefit from project management software as they track various activities in a project. Highlight eight benefits. **(8 Marks)**

**QUESTION TWO** **(20 Marks)**

- a) Explain how Robots are used in businesses today. **(2 Marks)**
- b) Describe two forms of encryption processes used by business to send secure information. **(4 Marks)**
- c) The design phase of the System Development Life Cycle (SDLC) is concerned with how the new information system will work. Highlight three types of design tools used by organizations today. **(6 Marks)**
- d) The enterprise architecture is the big picture for the organization. Identify and discuss four components of the enterprise Architecture. **(8 Marks)**

**QUESTION THREE** **(20 Marks)**

- a) Differentiate between Shielded Twisted Pair Cable and Unshielded Twisted Pair Cable **(4 Marks)**
- b) Give an illustration using a diagram of Client/Server database systems that minimize network traffic by dividing the processes between the client and server. **(8 Marks)**
- c) The extended value chain often incorporates contributions from buyers who help add value to the company's products or services. Describe its four support activities. **(8 Marks)**

**QUESTION THREE** **(20 Marks)**

- a) Differentiate between disaster recovery and business continuity plans. **(4 Marks)**
- b) i) Identify six steps and its related activities organizations take when they are deciding to buy rather than build software. **(6 Marks)**
- ii) Differentiate between Request for Information (RFI) and Request for Proposal (RFP) **(2 Marks)**

e) File management systems enable organizations to maintain its own records with its own computer files, each containing information that is required for operations, However, there are four problems that has surfaced as other offices began to develop their own file processing systems. Describe them. **(8 Marks)**

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